Digital preservation: some underlying issues for long-term preservation

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Introduction

The impact of digitization has been tremendous in recent years with the rapid growth of publications in digital format. In fact, libraries have been focusing on procurement of digital documents, given the fact that a digital document has varied accessibility and better resistance to regular wear and tear. Government of India has initiated a significant movement under their "Digital India" program, which aims to bring radical changes and exponential growth in society through Information Technology.

Advanced technologies for storage of digital information are abundant; however, technologies ensuring long-term preservation lag far behind. Digital preservation also preserves digital objects long-term in their original form. Digital media has proved its importance to preserve the information content of non-digital media such as traditional papers, paintings, photographs, musical recordings and movies. Several projects towards media conversion such as the Universal Digital Library of Carnegie Mellon University (www.ulib.org/), the International Dunhuang Project by the British Library (http://idp.bl.uk/), the Google Books Library Project, Project Gutenberg and Internet Archive, all share the mission of universal access to available knowledge. These initiatives lead the tremendous growth of digital information adding the new set of challenges for library and archives apart from their regular task of preserving their legacy materials in traditional formats.

The technological growth and its fast obsolescence have generated many concerns towards keeping the information contained in digital objects intact. Also, concerns are being shown towards sustainability to support both hardware and software (Ritschard, 1998). The digital media, made up of magnetic and optical devices, are more vulnerable to deterioration and fast obsolescence than their traditional counterparts. Hence, digital works created and stored using particular software and media would be susceptible to permanent loss due to lack of any future support from a software/media provider. Migration or conversion would take time and would be vulnerable to non-compliance with new emerging technologies.

Preservation and digital preservation

Preservation is an established subject and responsibility of libraries. Preservation is mostly defined based on the type of material to be preserved. Coordinating Council of Audiovisual Archive Associations (CCAAA) defines preservation as a process that is inclusive of all the steps that ensure permanent accessibility without any compromise with maximum integrity (CCAAA, 2005). In context to libraries and museums/archives, preservation is reducing the process of deterioration to ensure longevity and continued access. The majority of preservative actions are either preventive or curative. Preventive preservation is limited by environmental control and protection from undesired happenings. Curative preservation, on the other hand, aims to restore the object which is already damaged or endangered.

In digital preservation, there is no procedure done on the physical object (or analog object) rather a digital copy or digital surrogate of the object is prepared which stores all information content of the object including dimensions, coloration, etc. This digital surrogate can be used for study and easily transported over a network. The original object is required when only the physical properties of the material(s) of the object are to be studied, thus, allowing less exposure of the original object and increased longevity.

Apart from the digitization of the objects, there is exponential growth in publications of digital borne documents like e-books, e-journals and other electronic forms and formats, credit to the advent of electronic publication. Publishing is now no more restricted to printed text or digital text rather literature is also available as live events (as video), audio commentaries, graphics, etc. Hence, a range of documents for digital preservation has created a complicated situation to handle (Figure 1).

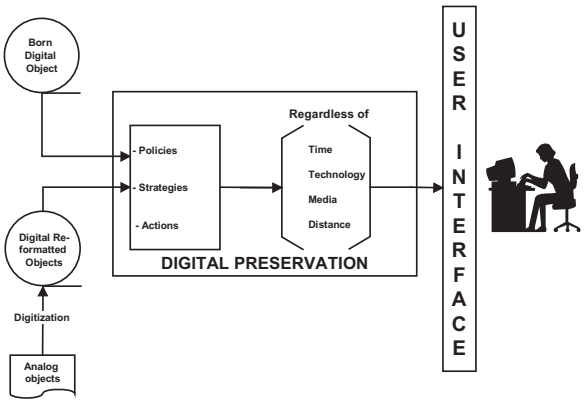
The scope of digital preservation is restricted not only to preserve the information (or the *bitstream*) contained in the object but also the storage media. Storage media is significant to achieve the full performance of the object. For example, an 8-bit program needs an 8-bit processor. Hence, for preservation of any 8-bit object (or program), we must at least preserve an 8-bit machine to play/execute the object (or program). The reason is advanced technology may diminish the utility and availability of these devices and none of the future devices, probably, would be able to play/execute the object (or program). Preserving storage media or hardware would involve traditional preservation techniques to keep it safe from humidity, temperature, light and the growth of biological objects.

Long-term digital preservation

Long-term preservation encompasses indirect efforts to increase the life expectancy of the objects (International Centre for the Study of the Preservation and Restoration of Cultural Property (ICCROM), 2001). The life expectancy of a digital object can be increased by specifying the storage and handling specifications along with the object itself.

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Figure 1. *Overview of digital preservation*



The specification may include preservation, descriptive and administrative information which will be crucial for the intended future use of the object. Some preventive preservation activities suggested for digital objects are control of temperature, humidity and lighting, packaging, preventing physical damage as well as assigning metadata for preservation, description and administration.

In curative preservation, the objective is to restore the damaged object in its original form. It involves remedial methods such as repairing, mending, fumigation, de-acidification, lamination and many others. The traditional approach for the restoration of artifact can be tedious and different curative activities such as cleaning, chemical treatment and repairing may be required. The objective of curative preservation is to retain the original form/format of the object.

In digital preservation, the focus is to restore the original bit-stream rather than retaining the storage media. The Cedars Project recommends media independence as one of the strategies for long-term preservation (CEDARS, 2002). However, some experts may not agree because it leads to detachment of the original bitstream from the original media. As such, any claim to restore the object in its original form without any data loss is highly unlikely.

There are, however, methods to treat digital objects to retain their originality such as migration and emulation strategies. Migration, as defined in the reports of The Commission on Preservation and Access and the Research Libraries Group (1996), is:

> [. . .] a set of organised tasks designed to achieve the periodic transfer of digital materials from one hardware/software configuration to another, or from one generation of computer technology to a subsequent generation.

Migration is a preservation strategy for a digital object to retain the object in its current version of file format, or to another, which is easier to handle and ensures continued access.

Open archiving information system (OAIS) breaks down migration into two categories, i.e. reversible and non-reversible. These processes ensure or evaluate the amount of information lost during data migration. As per OAIS, there are four strategies that fall within the purview of migration preservation strategy, ordered by increasing risk of information loss (Sawyer, 2005). These include refreshment (copying the bit-stream into a state of art storage or changed storage media), replication – bit-stream mirroring or object mirroring, repackaging – migration in which there is some change in the bits of the packaging information and transformation to preserve the full information content allowing some changes to the digital object.

Migration, as suggested by Wheatley (2002), depends on the level of technology involved. Minimum migration requires the least amount of technical work but may result in loss of database functionality and permanent data loss. Preservation migration aims to preserve the original look and feel of the digital object and is often cost-effective. Re-creation migration creates a new digital object with significant properties of the original and requires heavy financial investment. The new digital object does not incorporate any elements of the original; hence, there is a complete loss of functionality. Human conversion migration recreates the software elements of the digital object and reuses as much of the data of the original object as possible to ensure more accurate reproduction of the digital object. Automatic conversion migration converts the digital object to a current environment. The original environment is not preserved; so, there may be a loss of formatting and structure and also, the intellectual content. The process is to be repeated when the current format becomes obsolete associated with data losses.

Emulation is creation of a duplicate environment in which the original digital object is untouched. For example, WINE is a windows emulator for a Linux environment. Emulation, which was a bit underestimated in OAIS-2002, is now recognized as a viable preservation strategy (Sierman, 2012) in OAIS-2012. Emulation is advocated as the only significant approach to digital preservation by some scholars (Rothenberg, 1998). Others have a moderate opinion (Bearman, 1999; Bennet, 1997). Granger (2000) suggests deciding strategically on the method of digital preservation depending upon individual needs.

Known issues for long-term digital preservation

The object itself is the core for archiving, but an object has various facets which should be taken into consideration to ensure optimum performance of the archived object. In an archive, each object may have different stored versions and each requires a unique identifier such as the Uniform Resource Identifier. It would be appropriate to establish a global registry system and adhere to it. For an online environment, several systems like Digital Object Identifier, Handle Server and Uniform Resource Naming Systems, all provide options for unique

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identification. Local systems must also be developed considering individual/local requirements. With the standard registry system, an object can be universally identified or located and support interoperability among different persistent identifiers. Several frameworks (Bellini *et al.*, 2012; Kahn and Wilensky, 1995; Dunshire, 2007) have been proposed to achieve interoperability.

The archived object has to be encoded with administrative, descriptive and behavioral metadata at the beginning. A standard towards tailoring different metadata schemas as a single bundle attached to the object appears to be an ideal solution (Figure 2). The idea is the use of universal identification of the object to achieve platform independence and facilitate faster and continued access.

METS (Metadata Encoding and Transmission Standard) format is an XML (eXtensible MarkUp Language)-based standard for storage and transfer of metadata across archives. The METS format is independent to any metadata schema. Hence, different metadata schemas can be used with METS. The used schemas must ensure: identification of objects and use of persistent identifiers, object description, access control, ownership, object behavior or attributes and object retrieval.

METS also has the facility to incorporate preservation metadata. Under administrative metadata section <amdSec> the element <digiprovMD> is used for recording digital preservation information. For digital preservation, a locally tailored schema or standard schema like PREMIS can be used. PREMIS, sponsored by Library of Congress, is a tool for developing a preservation infrastructure for long-term management of digital objects (Guenther, 2010):

```
<mets:amdSec>
  <mets:digiprovMD ID=
   "DP_0755ad93-5fd1-11da-
    b211-19e7a5cf4814"
  CREATED="2006-11-
  27T21:37:13">
  <mets:mdWrap MDTYPE=
  "PREMIS">
    <mets:xmlData
    xmlns:premis = www.loc.
    gov/standards/premis/v1>
 <mets:xmlData>
      <!—using PREMIS with
          namespace for digital
          preservation–>
  </mets:xmlData>
 </mets:mdWrap>
</mets:digiprovMD>
</mets:amdSec>
```

But again, interoperability remains the key issue while using different metadata schemas.

Whether working in a traditional or digital environment, intellectual property rights are should be used while planning for long-term preservation. Deegan and Simon (2006) has listed several measures to be undertaken concerned to copyright. These include: identifying the circumstances for which digital preservation is to be undertaken, obtain the permission from the content creator for copying, storing or any other activity for archival purpose, identify the circumstances the content being preserved be made accessible in the online/digital environment and to identify the metadata required to record all details regarding the rights holders and the collection.

A suitable, sustainable and scalable storage medium for a digital archive is required to manage vulnerability to rapidly evolving technology. An objective approach to select a suitable storage medium depends on the size, period of storage and data readability and speed. The durability of the medium is also vital for selection of storage medium. Nevertheless, new improved technology may always outlast the storage devices or media to store or read the data. With quick obsolescence of technology.Suitable file storage formats have become a major issue in digital archiving across the world. There are three types of file formats:

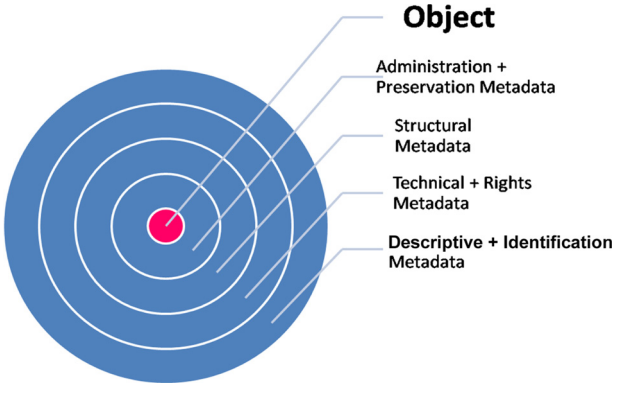
1. proprietary with closed specifications e.g. doc, xls and psd;
2. proprietary with open specifications e.g. pdf, rtf and tiff; and
3. non-proprietary with open specifications e.g. html, otd, png and jpeg and election of non-proprietary specifications with specialized applications that usually means using proprietary specifications requiring negotiations with the vendor.

For the storage of graphics, the selection between raster and vector graphics should be used. Presently, vector graphics lack any non-proprietary and open specification until new specifications become available. There are no specific guidelines to select the appropriate format for archival preservation. Rapidly changing technology will ensure better and advanced means of representation and storage.

Each process already executed or required to be done, must be documented including the state to which the object has been brought in for archiving. The following steps must be taken to ensure quality control in archiving:

- recording the condition of the object during the process of archiving, documentation, testing object semantic and syntactic values, of the content of the object;
- establishing a procedure manual and continual review, recording references to the related objects or

Figure 2. *Bundling object and metadata*



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environment or version control information; and

- establishing methods of testing and periodic review of object stock and ingestions in the archive and calibration of different equipment used in the preservation process, such as scanner, OCR software and digital content writers/readers.

Digital archives store the precious knowledge of the past and hold a great value for today's information society. Hence, there should be a plan to deal with any risk or threat (natural or man-made). For this, data could be either be centrally stored or distributed. In a centrally located storage, data recovery will be difficult at times but in case of any disturbance, there are chances of hampering. In a distributed system, an archive is spread over different geographical locations as sub-archives (see Figure 3). The queries in such systems would get transferred to suitable sub-archives with the help of an intermediary service provider so that any disturbance in one location will remain safe at other locations also known as offsite mirroring. Apart from mirroring, suitable authentication methods should be implemented for authentication of users (see Figure 3).

Disaster planning and risk preparedness is vital in the digital environment. Some of the famous systems such as LOCKSS (Lots of Copies Keeps Stuff Safe) and iRODS (Integrated Rule Oriented Data Systems) aimed long-term digital preservation with a built-in mechanism for disaster planning (Frank, 2017). A possible checklist for risk preparedness in case of digital archives may include: the dentification of potential, moderate and remote risks, forming a disaster management team and communication channels, security such as firewalls, automatic data recovery through RAID, backup with mirroring and digital certification and password encryption.

Concluding remarks

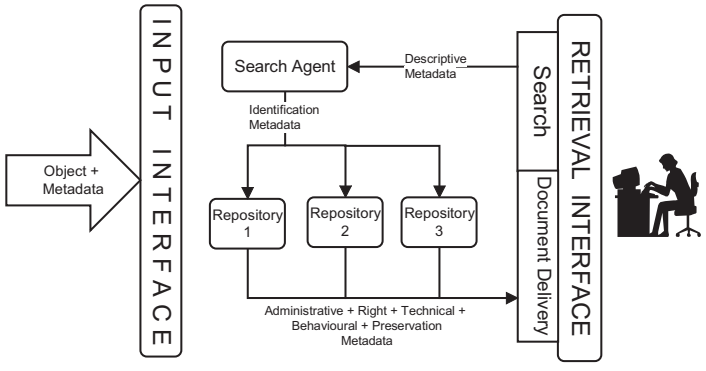
With born-digital, documents have created new challenges in keeping the record in original form with established authenticity over a period of time. Social and institutional scenarios may also be the deciding factor for the perpetual maintenance of a digital archive (Altman *et al.*, 2009).

Detachment, data migration (Roedig *et al.*, 2003) and emulation are some of the tested measures against losses of digital information. However, any intermediary procedure must not change the operation of the object from the original. Every step towards the processing must be recorded for future use and any procedures applied on the object has to be reversible. For physical objects, there are methods to reverse the process of archival but the technological advancement for the preservation of digital object is yet to be achieved.

Currently, there are no set rules or standards for digital preservation. OAIS model is a mix of management and technical functionalities making it difficult to develop a system that is more independent of the managerial flip-flops (Egger, 2006). There are only use cases that have set forth some laid procedures for long-term preservation. These documents share their experiences of handling projects and provide an overview of dealing with a digital preservation initiative.

Libraries and Archives need to plan their preservation programs in an era of rapidly evolving technology and limited funding for management. Existing standards such as OAIS, PREMIS and the outcomes of various digital preservation projects should be helpful in developing a framework for digital preservation for at least several years.

Figure 3. *Searching among distributed archives*



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